

Project IT362 Report – Phase#2

-Group member-

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Introduction

This report presents the results of the Exploratory Data Analysis (EDA) conducted for both primary and secondary datasets related to the Saudi Labor Law. The purpose of this analysis is to better understand the content and structure of various legal and public data sources that reflect different perspectives on labor regulations in Saudi Arabia.

The analysis aims to identify key patterns, relationships, and differences across four main data sources: the official labor law from the Board of Experts (BOE), public consultation data from the Istilaa platform, explanatory articles from the Qiwa platform (Qiwa Articles), and user-generated questions from the Qiwa FAQ section. These datasets together provide a comprehensive view of how labor law information is presented, interpreted, and discussed across official and public platforms.

This EDA was performed after completing the data processing and cleaning phase to ensure the accuracy and reliability of the findings. A combination of descriptive statistics, textual visualizations, and comparative charts was used to uncover meaningful trends and thematic similarities between the datasets.

The results of each dataset were analyzed and presented separately to highlight their unique characteristics, potential biases, and target audiences. Finally, the report includes a comparative overview and a summary of the main insights and new hypotheses derived from the analysis.

Primary Data: BOE

Brief description:

The dataset was obtained from the Saudi Council of Experts at the Council of Ministers, a specialized advisory body responsible for preparing, reviewing, and amending laws and regulations in Saudi Arabia. It includes the latest version of the Saudi Labor Law with all its articles and amendments.

Dataset overview:

```
◆ BOE Dataset Info:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 248 entries, 0 to 247
Data columns (total 6 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Section     248 non-null   object
1   Chapter     248 non-null   object
2   Article     248 non-null   object
3   Text        248 non-null   object
4   Status      248 non-null   object
5   Source      248 non-null   object
dtypes: object(6)
memory usage: 11.8+ KB
```

Figure 1

Status distribution in BOE:

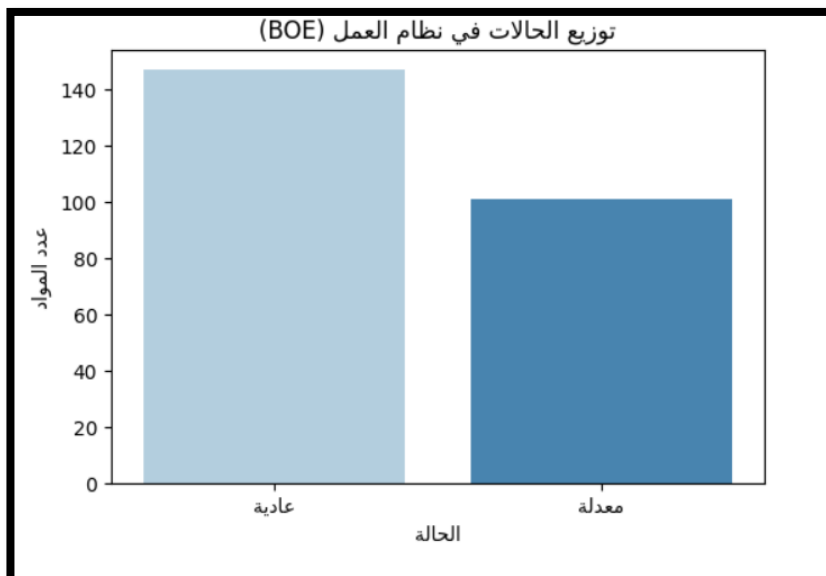


Figure 2

The most frequent sections in the BOE Labor Law:

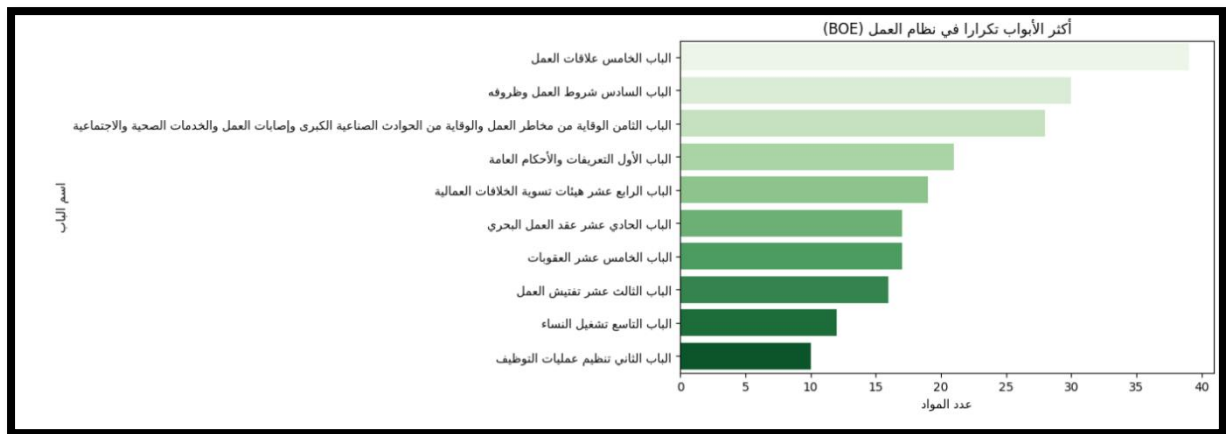


Figure 3

Relationship between section and article status:

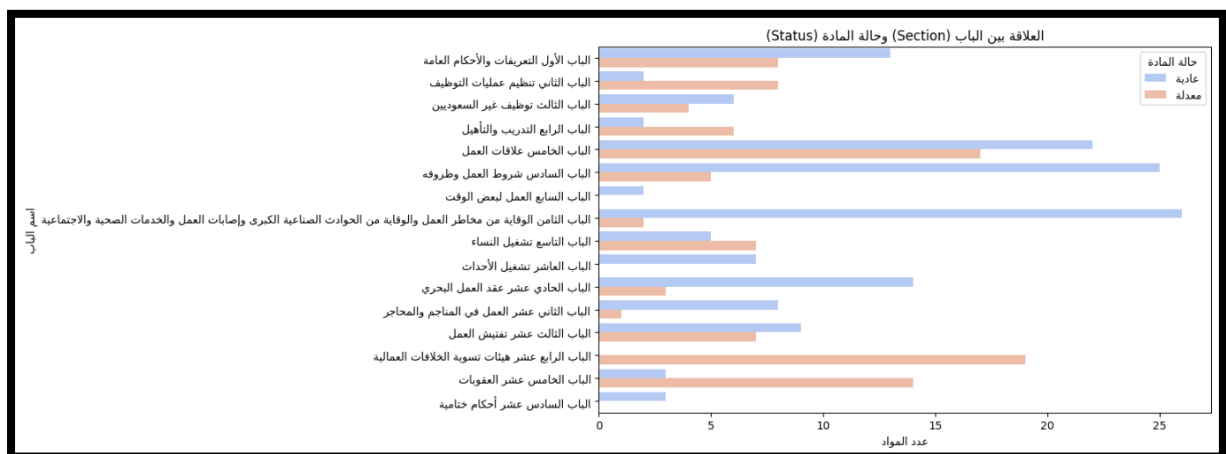


Figure 4

BOE Insights:

The BOE dataset reveals that legislative modifications are concentrated within specific sections of the Saudi Labor Law. Certain sections show a higher frequency of updated or amended articles, indicating that regulatory efforts primarily focus on a limited set of labor-related themes rather than being uniformly distributed across all topics.

Secondary Data:

A-First Source: Istitlea

Brief description:

The dataset was collected from the Saudi “Isttelaa” platform, which gathers public and institutional feedback on draft laws and regulations. It focuses on the latest Saudi Labor Law.

Bias:

Although the data is sourced from an official government consultation platform, the analyzed texts are public opinions, comments, and suggestions rather than binding legal texts. This introduces an Opinion Bias, meaning the content includes a high degree of personal judgment and estimations which the project aims to minimize in the new regulations and penalties table.

Dataset overview:

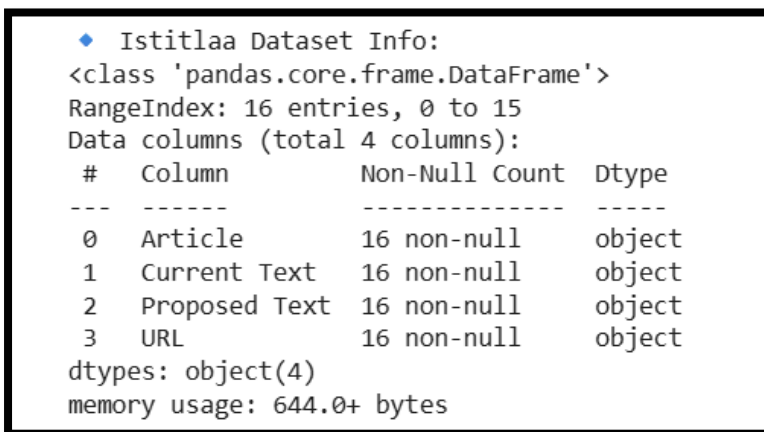


Figure 5

Article proposal frequency:

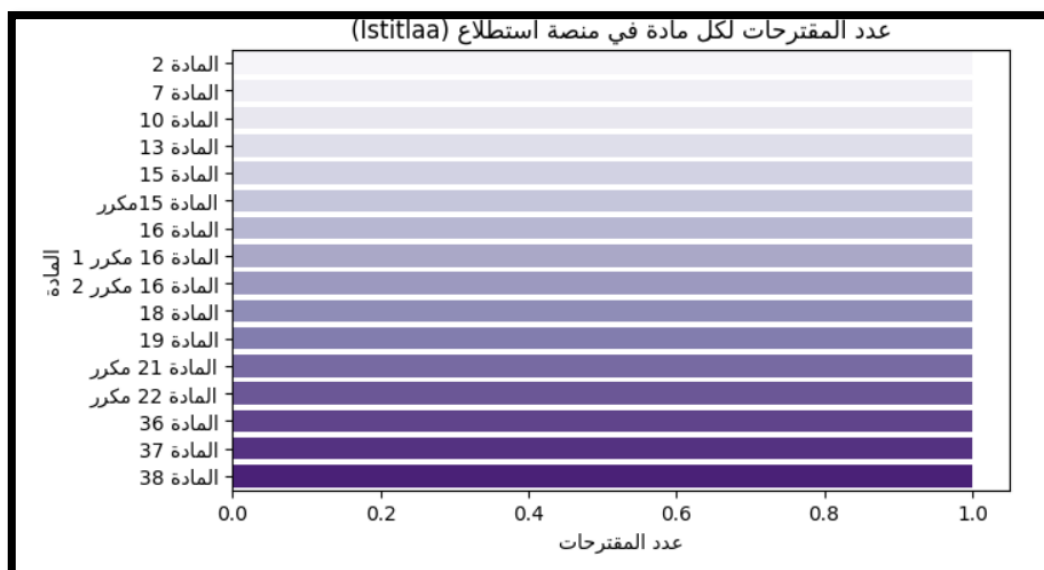


Figure 6

Categorize proposed changes:

	Article	Change_Type
0	المادة 2	حذف المادة
1	المادة 7	تعديل أو استبدال النص
2	المادة 10	تعديل أو استبدال النص
3	المادة 13	تعديل أو استبدال النص
4	المادة 15	تعديل أو استبدال النص

Figure 7

Distribution of change types:

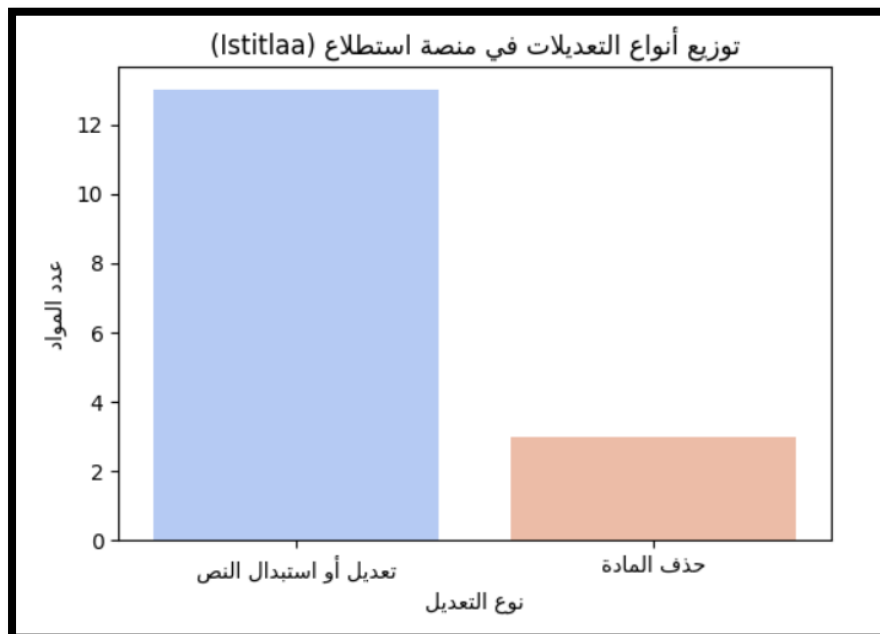


Figure 8

Istitlaa Insights:

The Istitlaa dataset demonstrates that most public feedback emphasizes practical and actionable changes—such as requests for updates, training policies, or procedural modifications. The frequency distribution of proposed changes highlights that public participants focus on improving clarity and applicability in specific areas rather than suggesting broad or theoretical amendments.

B-Second Source: Qiwa articles

Brief description:

The dataset focuses on the Saudi Labor Law from the “Qiwa” platform, covering its articles and related explanatory materials for employers. While the platform offers additional tools and examples, this dataset specifically includes the core labor law content.

Bias:

These articles are designed as comprehensive and educational texts targeting employers, which introduces an Interpretation Bias toward presenting an idealized, systematic view of the labor law. They may not sufficiently cover the specific details of violations and penalties which is the core focus of this project's updated table. The content, therefore, requires supplementation with data that offers more concrete specifications.

Dataset overview:

```
=== Qiwa (Cleaned) ===  
Shape: (16, 2)  
  
Columns:  
['URL', 'Content']  
  
Data types:  
URL      object  
Content  object  
dtype: object  
  
Missing values per column:  
URL      0  
Content  0  
dtype: int64  
  
Approx. duplicate rows: 0
```

Figure 9

Content length distribution:

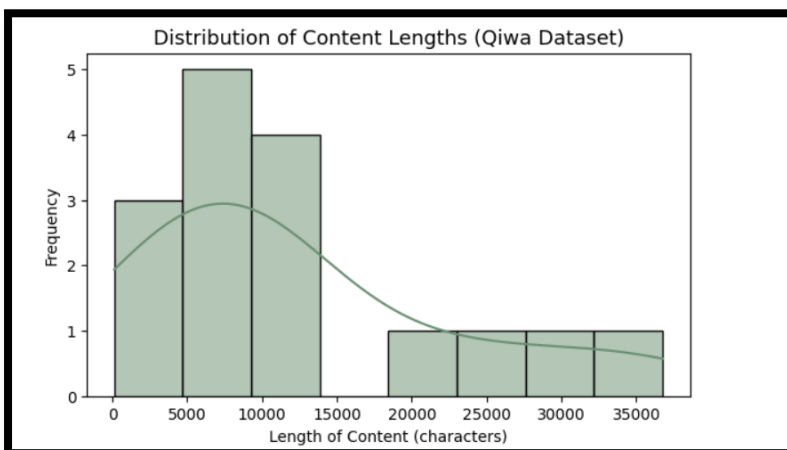


Figure 10

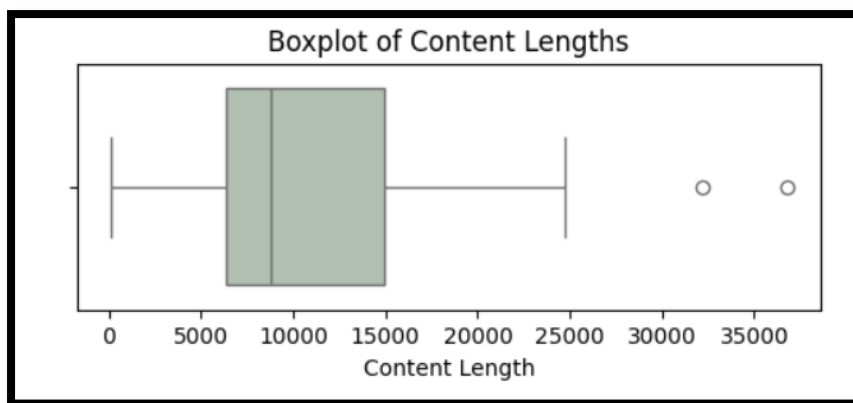


Figure 11

Top frequent words:

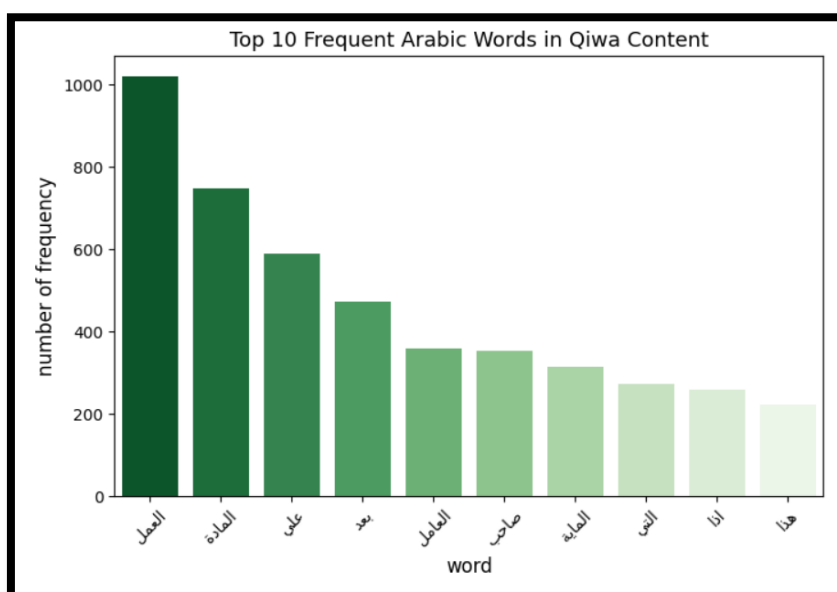


Figure 12

Word cloud:

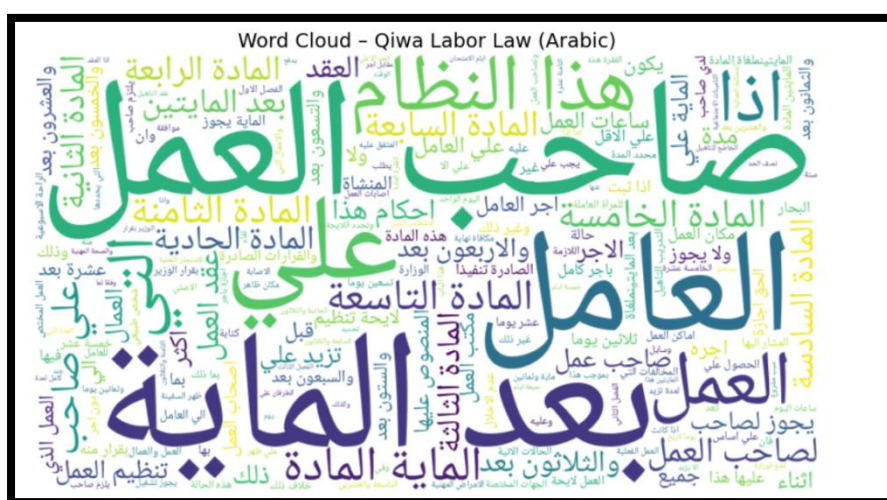


Figure 13

Qiwa Articles Insights:

The Qiwa Articles dataset contains lengthy and detailed texts that reflect a formal, explanatory tone. The frequent appearance of legal and procedural terms indicates that this content serves an educational purpose, offering systematic interpretations of labor law articles rather than discussing penalties or real-world applications.

C-Third Source: Qiwa FAQ

Brief description:

The FAQ part of Qiwa platform.

Bias:

This data reflects the most common questions raised by Qiwa platform users. Consequently, it carries a Usage Bias, focusing heavily on routine platform usage and practical daily issues, potentially neglecting less common or highly complex legal topics relevant to the broader labor law.

Dataset overview:

```
=== FAQ (Labor Law FAQ) ===
Shape: (16, 2)

Columns:
['Question', 'Answer']

Data types:
Question    object
Answer      object
dtype: object

Missing values per column:
Question    0
Answer      0
dtype: int64

Approx. duplicate rows: 0
```

Figure 14

Distribution of question lengths:

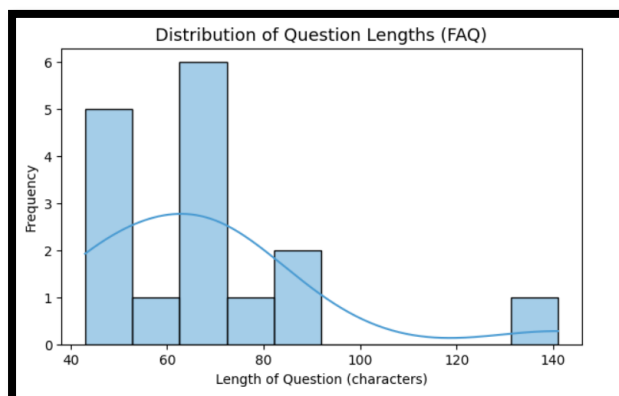


Figure 15

Top frequent words:

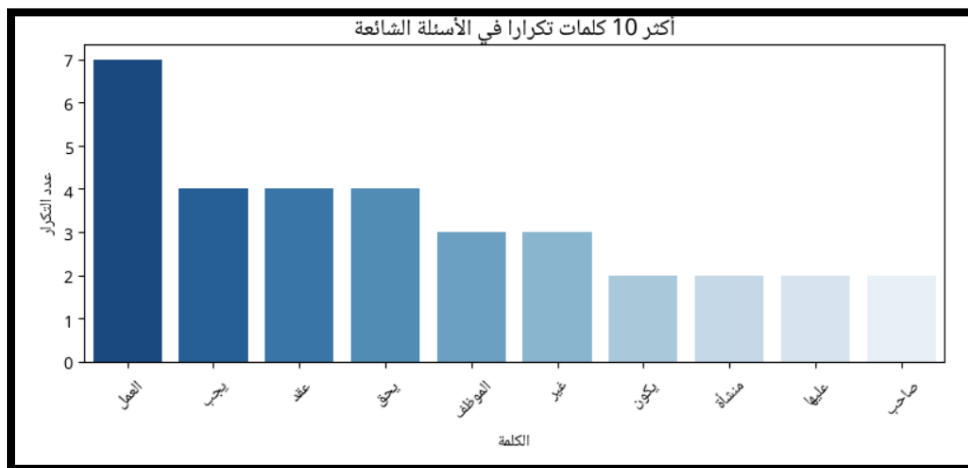


Figure 16

Word cloud:

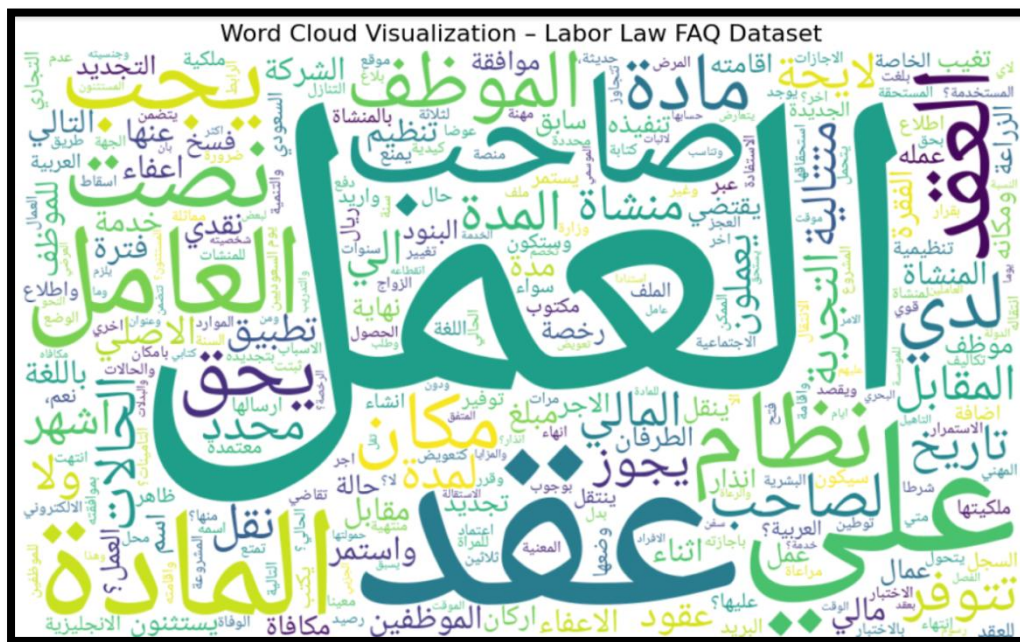


Figure 17

N-gram analysis bigrams & trigram:

Top 10 Bigrams:		
	Bigram	Count
136	هل يحق	4
73	عقد العمل	3
154	يجب أن	3
107	ما هي	3
2	أن تتوفر	2
23	التي يجب	2
100	لصاحب العمل	2
70	صاحب العمل	2
54	تتوفر في	2
92	في عقد	2

Top 10 Trigrams:		
	Trigram	Count
1	أن تتوفر في	2
22	التي يجب أن	2
149	يجب أن تتوفر	2
86	في عقد العمل	2
130	هل يحق لصاحب	2
152	يجب لصاحب العمل	2
50	تتوفر في عقد	2
3	أن ينقل موظف	1
0	أخرى دون موافقة	1
9	إلى غير محدد	1

Figure18

Histogram of answer lengths:

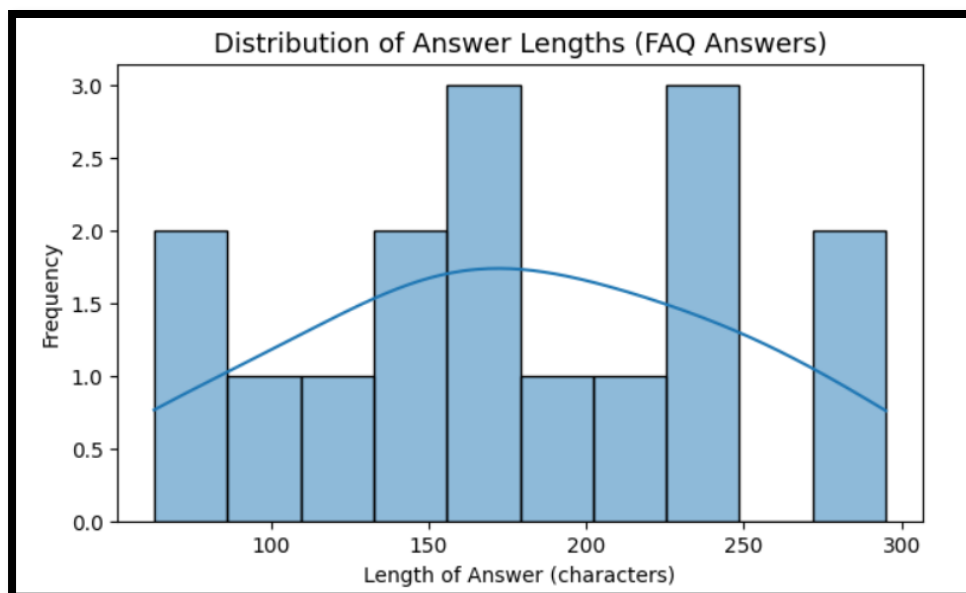


Figure 19

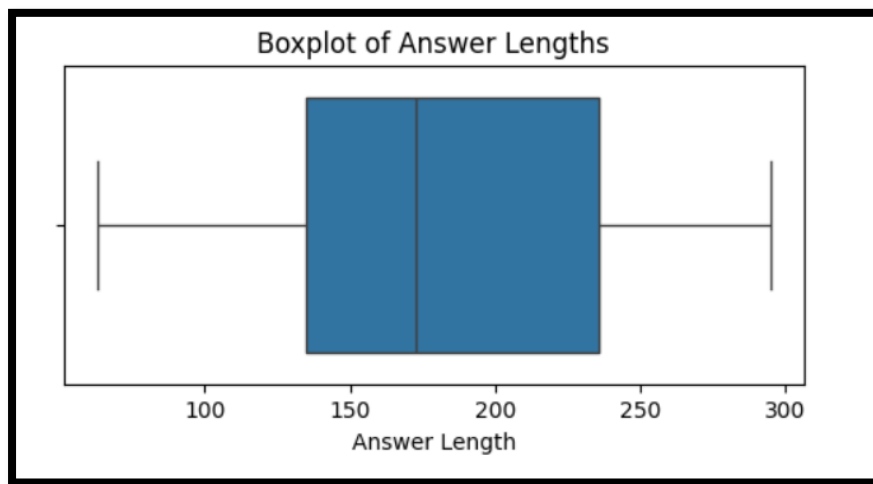


Figure 20

Qiwa FAQ Insights:

The Qiwa FAQ dataset consists of short, focused entries designed to address common user inquiries. The frequent terms and n-gram patterns suggest a strong emphasis on day-to-day employment issues and practical user concerns, showing that this dataset captures real usage patterns and frequently asked questions by workers and employers.

Comparison:

Comparison Overview:

We compared our four datasets (Qiwa FAQ, Qiwa Articles, BOE Official Law, and Istitlaa Public Consultation) based on text length, topic frequency, and content structure.

Interpretation of table Comparison:

The comparison across the four datasets reveals major differences in both text length and topic focus.

Qiwa (FAQ) has 32 entries with short texts (average 122 characters), which is consistent with its Q&A structure providing direct, concise information.

Qiwa (Articles) shows extremely long texts (average 12,262 characters, median 8,771), reflecting detailed explanatory content written in full-article form.

BOE (Official Law) includes 248 entries with moderate text length (average 426 characters), as it represents segmented legislative articles.

Istitlaa (Public Consultation) contains 16 entries with medium-length proposals (average 522 characters), showing a balance between brevity and detail for public consultation.

In terms of topics, all datasets share the recurring word (work), emphasizing labor-related content. However, Qiwa and BOE show a strong dominance of legal and procedural terms such as (article), while FAQ focuses on practical instructions and employee rights. Istitlaa highlights consultation-oriented terms like (training) and (modification), aligning with its goal of collecting public input on legal amendments.

Overall, the results show a natural hierarchy from concise user-focused content (FAQ) to highly detailed official texts (Qiwa), reflecting the nature and purpose of each platform.

	Dataset	Rows	Avg_Text_Length	Median_Text_Length	Top_Words
0	FAQ (Q+A)	32	122.88	87.0	العمل (7); يجب (4); عتد (4); علي (4); يحق (4)
1	Qiwa (Articles)	16	12262.50	8771.5	العمل (946); المادة (738); علي (587); بعد (470)
2	BOE (Official Law)	248	426.79	293.0	العمل (387); علي (339); بعد (292); المايئين (2)
3	Istitlaa (Public Consultation)	16	522.44	300.0	التدريب (41); المادة (36); العمل (28); علي (21)

Figure 21

Interpretation of chart Comparison:

The chart shows a large difference in median text length across datasets. Qiwa has a significantly higher median due to its full-length articles, while FAQ, BOE, and Istitlaa contain much shorter texts. This confirms that Qiwa provides detailed explanatory content, whereas the other datasets focus on more concise, structured information.

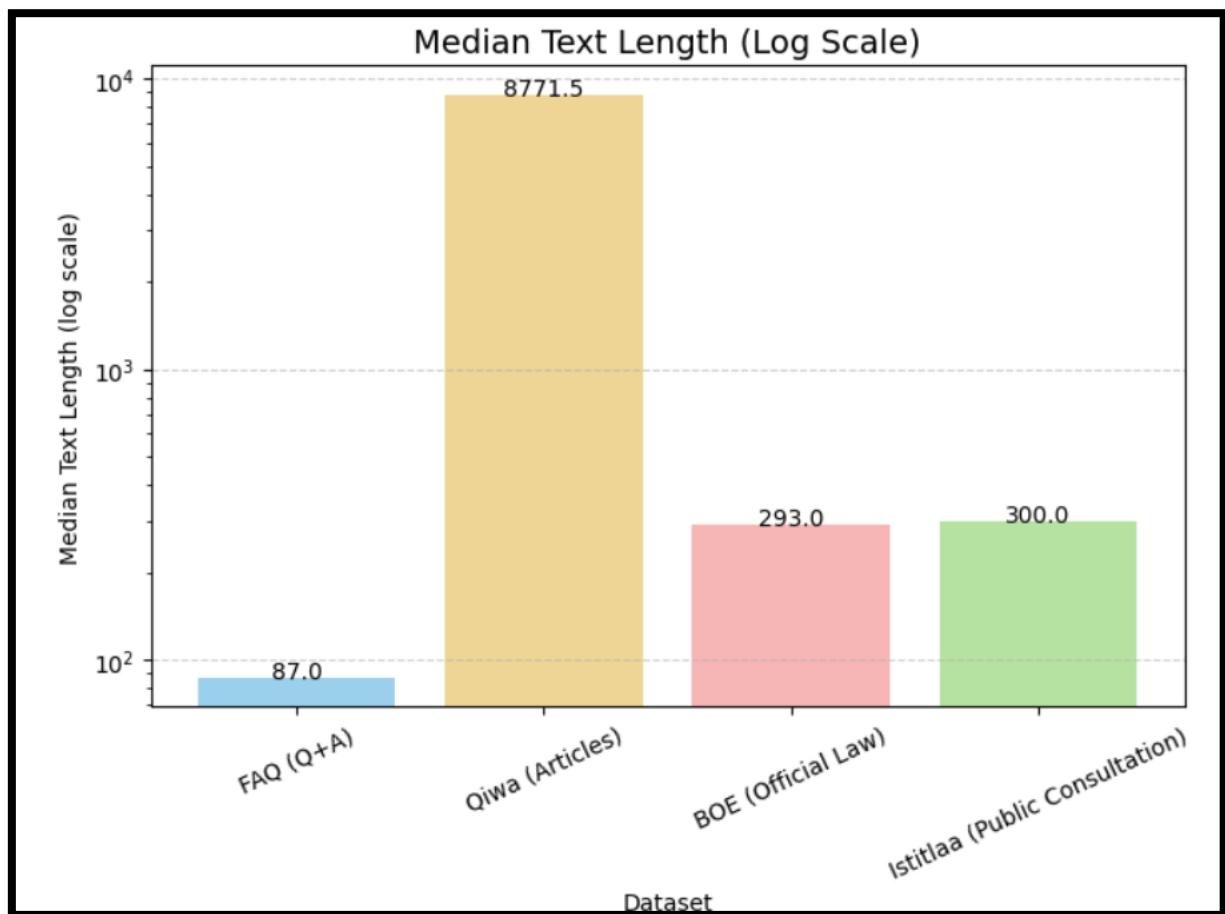


Figure 22

Summary of new insights and hypotheses:

New Insights:

- The datasets vary significantly in text length and content structure, reflecting their distinct purposes.
- **Qiwa (FAQ)** provides concise, user-oriented content designed for quick information retrieval.
- **Qiwa (Articles)** contains much longer and more formal texts, indicating its role as an official platform for detailed regulations.
- **BOE (Official Law)** includes moderately long legal articles, presenting structured and segmented legislative content.
- **Istitlaa (Public Consultation)** shows medium-length texts focused on proposed changes and public feedback.

Common topics across all datasets revolve around labor laws, regulations, and rights, emphasizing the shared domain of employment and governance.

However, the frequency and depth of certain terms (like “Article” and “work”) highlight how each platform targets a different audience — from general users to policymakers.

Hypotheses:

- The difference in text length may correlate with the target audience, shorter texts for the public (FAQ) and longer ones for legal experts (Qiwa, BOE).
- The dominance of certain keywords (e.g., “work”) suggests a strong thematic overlap, indicating potential for cross-dataset topic alignment in future analysis.
- Public consultation texts (Istitlaa) may serve as a bridge between official laws (BOE) and user interpretations (FAQ/Qiwa).